Project Multivariate and Hierarchical Data – Sample Size Exercise

You were contacted by the R&D discovery department of your company to help them to design their experiment regarding a potential antiviral for an infectious disease. They typically conduct simple experiments: a cotton rat is provided a daily dosing of a test drug starting on day 1 until day 7, next on day 3 the cotton rat is challenged with the infectious disease, and then on day 7 the viral titer in the lungs is measured. For a good antiviral it is expected that the viral titer is lower.

They provided you with information below and they expect from you an initial suggestion that would be a starting point for further discussion. Therefore, you can feel free to set some parameters that would otherwise be set by a domain expert. No other predictors play a role in this experiment.

Your task is to prepare an initial sample size calculation. That means to specify all the five points that we saw in the lecture and then compute the sample size.

The experiment description:
They usually follow a standard design in which a cotton rat is provided a daily dosing of a study drug starting on day 1 until day 7, next on day 3 the cotton rat is challenged with the infectious disease, and then on day 7 the viral titer in the lungs is measured. In this experiment we will compare two antivirals: (1) the standard antiviral TK35 used as the standard of care for this disease; (2) and our experimental antiviral PYV123. The expectation is that our antiviral PYV123 reduces the viral titer in the lungs compared to TK35. We want to know how many cotton rats we should include in our experiment to make sure we can draw useful conclusions. Usually, we use 4 cotton rats per group.

In case the experimental antiviral could decrease the viral titer in the lungs with 1.5 or more ($\log_{10}$ PFUe/g) this would be considered scientifically relevant. From previous research, it is well known that the viral titer measurements follow a normal distribution. Using a similar study design, we already have data available on TK35 in 6 cotton rats. The viral titer measurements are: 4.6 6.2 5.9 5.3 6.3 4.4

Our question is: Would n=4 per group be sufficient? If not, what should be our sample size to fulfil aforementioned goals?

Hint: to compute the sample size
Just replace **x** in following codes for R or SAS with appropriate value. Note that you need to leave empty the option for sample size setting (*n* in R and *npergroup* in SAS) in order to get result.

R:
```
power.t.test(n = , delta = x, sig.level = x, sd = x, power = x, type = "x", alternative = "x")
```

SAS:
```
proc power;
twosamplemeans test = diff
groupmeans = x|x
stddev = x
alpha = x
sides = x
npergroup = .
power = x ;
run;
```